

Mathematical Model for Physarum Lab

Interactive C++/WebAssembly slime mold simulator

May 23, 2026

1 Introduction

Physarum Lab is an idealized computational simulator inspired by *Physarum polycephalum*. It combines a Jones-style particle and trail model [Jones, 2010], an energy and growth budget, and a low-frequency adaptive nutrient-network overlay motivated by the Tero transport model [Tero et al., 2007]. It is not a calibrated biological model; its purpose is interactive exploration of qualitative mechanisms: chemotaxis, food depletion, growth, search/exploit switching, and path reinforcement.

2 State and Notation

The domain is a rectangular grid

$$\Omega = \{0, \dots, W - 1\} \times \{0, \dots, H - 1\},$$

with continuous agent positions embedded in $[0, W) \times [0, H)$. Agent i has

$$X_i(t) = (p_i(t), \theta_i(t), E_i(t), m_i(t), \mu_i(t), a_i(t)),$$

where $p_i = (x_i, y_i)$ is position, θ_i is heading, E_i is internal energy, m_i is biomass, $\mu_i \in \{\text{search, exploit, dormant}\}$ is behavior mode, and a_i is alive status.

Food source k has

$$F_k(t) = (f_k, C_k(t), q_k, \sigma_{A,k}, r_k),$$

where f_k is position, C_k is remaining calories, q_k is quality, and r_k is the eating radius. Trail and food fields are $T_t(x, y)$ and $A_t(x, y)$.

3 Food Attractant and Trail

Active food creates a saturating attractant field:

$$A_t(x, y) = \sum_k q_k \frac{C_k(t)}{C_k(t) + C_{1/2} + \varepsilon} \exp\left(-\frac{\|(x, y) - f_k\|^2}{2\sigma_{A,k}^2}\right).$$

Agents deposit trail by bilinear splatting:

$$D_t(x, y) = \sum_{i:a_i=1} q_0 m_i(t) \phi_E(E_i) \phi_\mu(\mu_i) K_h((x, y) - p_i(t)),$$

where $\phi_E(E) = \text{clip}(E/E_{\max}, 0, 1)$ and

$$\phi_\mu(\mu) = \begin{cases} 1, & \mu = \text{search}, \\ 1.25, & \mu = \text{exploit}, \\ 0.15, & \mu = \text{dormant}. \end{cases}$$

Trail evolves by diffusion and decay:

$$T_{t+\Delta t} = (1 - \lambda_T \Delta t) (G_{\sigma_T} * (T_t + D_t)).$$

The implementation approximates G_{σ_T} with local blur passes for interactive speed.

4 Sensing and Motion

The combined sensing field is

$$S_t(x, y) = w_T T_t(x, y) + w_A A_t(x, y) - w_R R_t(x, y) - w_O O(x, y).$$

Let $u(\theta) = (\cos \theta, \sin \theta)^T$. Front, left, and right sensors are

$$z_i^\delta(t) = p_i(t) + d_s u(\theta_i(t) + \delta \alpha), \quad \delta \in \{-1, 0, +1\}.$$

The agent samples $S_i^\delta = \mathcal{I}[S_t](z_i^\delta)$ using bilinear interpolation. It turns toward the strongest sensor:

$$\Delta \theta_i^{\text{det}} = \begin{cases} +\beta, & S_i^{+1} = \max_\delta S_i^\delta, \\ -\beta, & S_i^{-1} = \max_\delta S_i^\delta, \\ 0, & S_i^0 = \max_\delta S_i^\delta. \end{cases}$$

5 Search, Exploit, and Dormancy

The probability of continuing exploratory search is

$$P_i^{\text{search}} = \sigma \left(k_E \left[\frac{E_i}{E_{\max}} - \theta_d \right] \right) (1 - \sigma(k_S[S_i^{\max} - \tau_S])),$$

clipped to $[0, 1]$. High enough energy and weak local signal encourage search; strong trail or food signal encourages exploit mode. If energy is very low and the signal is weak, the agent enters dormant mode.

Food direction also biases heading:

$$F_i^{\text{food}} = \sum_k \frac{\chi(C_k > 0) c_k}{\|f_k - p_i\|^2 + \varepsilon} \frac{f_k - p_i}{\|f_k - p_i\| + \varepsilon},$$

$$\rho_i = \text{clip} \left(\|F_i^{\text{food}}\| \left(1 - \frac{E_i}{E_{\max}} \right)^\eta, 0, 1 \right).$$

The heading is blended toward $\arg(F_i^{\text{food}})$ by ρ_i , so hungry agents are more food-directed.

6 Energy, Eating, Growth, Division, Death

Energy loss per step is

$$L_i = \Delta t (c_b m_i + c_m v_i^2 m_i + c_s \mathbf{1}_{\mu_i=\text{search}}).$$

If $\|p_i - f_k\| < r_k$, the food gain is

$$G_{ik} = \min\{C_k, \eta_{\text{eat}} q_k \Delta t\}, \quad C_k(t + \Delta t) = C_k(t) - \sum_i G_{ik}.$$

Agent energy updates as

$$\hat{E}_i(t + \Delta t) = \text{clip} \left(E_i(t) + \eta_E \sum_k G_{ik}(t) - L_i(t), 0, E_{\max} \right).$$

Growth occurs above an energy threshold:

$$\Delta m_i = r_g \left[\frac{\hat{E}_i}{E_{\max}} - \theta_g \right]_+ m_i \Delta t, \quad E_i(t + \Delta t) = \hat{E}_i - \kappa_g \Delta m_i.$$

Agents divide when $m_i > m_{\text{split}}$ and $E_i > E_{\text{split}}$. Sustained zero energy for τ_{dead} steps causes death.

7 Adaptive Nutrient Network

At a lower update frequency, the simulator samples graph nodes from active food and trail ridges. Edges connect nearby nodes. Each edge has length ℓ_e and conductance D_e . The motivating continuous model uses

$$C_e(t) = \frac{D_e(t)}{\ell_e + \varepsilon}, \quad L_D = B \text{diag}(C_e) B^T, \\ L_D(t) P(t) = b(t), \quad Q_e(t) = C_e(t) (P_u(t) - P_v(t)),$$

with the Tero update

$$\dot{D}_e = |Q_e| - D_e.$$

The interactive implementation uses a bounded low-frequency variant: edges on the current nutrient path are reinforced and all others decay,

$$D_e(t + \Delta t) = \max \left(D_{\min}, D_e(t) + \Delta t \left[\alpha_D \frac{|Q_e|^\gamma}{|Q_e|^\gamma + q_0^\gamma + \varepsilon} - \mu_D D_e(t) \right] \right).$$

The displayed path uses Dijkstra cost

$$c_e^{\text{path}} = \frac{\ell_e}{(D_e + \varepsilon)^\eta}.$$

Transport cost and dissipation are reported as

$$J_{\text{transport}} = \sum_e \ell_e D_e, \quad \mathcal{E}_{\text{diss}} = \sum_e \frac{\ell_e}{D_e + \varepsilon} Q_e^2.$$

8 Numerics and Limitations

The default browser grid is lower resolution than the display canvas and is scaled for rendering. This keeps interactions responsive while preserving the qualitative trail dynamics. The current WebAssembly hook mirrors the C++ API; when Emscripten is not available, the TypeScript fallback runs the same state variables and equations for local development.

References

- Jeff Jones. Characteristics of pattern formation and evolution in approximations of physarum transport networks. *Artificial Life*, 16(2):127–153, 2010. doi: 10.1162/artl.2010.16.2.16202.
- Atsushi Tero, Ryo Kobayashi, and Toshiyuki Nakagaki. A mathematical model for adaptive transport network in path finding by true slime mold. *Journal of Theoretical Biology*, 244(4):553–564, 2007. doi: 10.1016/j.jtbi.2006.07.015.